

QVF-Hybrid: A Guaranteed-Floor Hybrid Quantum–Classical Architecture

Rigorous controls and an honest paradigm comparison on DeepLense

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The contribution in one slide

We built the strongest hybrid we could for the quantum side, ran it through a **complete control ladder**, and report exactly what survives.

Tier	Control	Verdict
Module	vs. matched <i>sham</i>	significant at (II, 250): $n=20$, $p=0.0032 < \alpha=0.01$; small (+1.4%, under our 3% practical bar)
Model	vs. <i>plain CNN</i>	guaranteed floor by construction; vs.-plain contrast not significant ($p=0.088$)
Paradigm	vs. <i>MAE-ViT</i>	significant (Holm $p=0.003$, 10/10, +22–28 pp), 1/19 params, zero pretraining data

We do *not* claim quantum advantage.

The problem: a baseline crisis in QML

Published hybrid quantum–classical wins routinely evaporate once the classical control is made honest.

- **Bowles et al. 2024**: many hybrid QNNs lose to *trivially small* classical baselines once architecture and parameters are matched.
- **Schnabel et al. 2025** (~20 000 models): no robust quantum edge.
- Our companion QOVT study: orthogonal (RBS) attention *matches but does not beat* classical multi-head attention.

Two failure modes

The quantum model is compared against either (i) a much **larger** classical model, or (ii) a **handicapped sham** whose deficit the circuit merely recovers.

Our response: a control *ladder* — sham *and* null baseline *and* multi-seed *and* pre-registration.

Task and data

Task. Classify 64×64 lensing images: **axion** (vortex) / **CDM** (subhalos) / **no_sub** (smooth).
3-class macro-OVR AUC, 9:1 split, best-val-AUC checkpoint.

Dataset	Character	Role
Model_I	lower SNR, ceiling ≈ 0.97	second dataset
Model_II	higher SNR, ceiling ≈ 0.99	primary signal dataset
Model_III	higher-SNR sibling	generalization check
CommonTest	GoogleSC vort/sphere/no	out-of-family <i>hard case</i>

The regime that matters. LSST / Euclid / Roman give many *unlabeled* images but few *labeled* ones. We test down to $N=100$ /class — where the classical SOTA is weakest.

Target. MAE-pretrained ViT, AUC 0.968 (arXiv:2512.06642); our protocol-matched reproduction 0.9895 (Model_II, full data).

Evolution (1): QVF-Scratch — and its apparent win

CNN $\rightarrow$ *Neural Amplitude Encoding* $\rightarrow$ 8-qubit StronglyEntanglingLayers ($T=96$)
 $\rightarrow \langle Z \rangle \rightarrow$ linear head. End-to-end, no pretraining.

Dataset	Quantum (142.8k)	Sham (144.8k)	MAE-ViT (2.72M)	Q-Sham
Model_I	0.9805	0.9790	0.9778	+0.0015
Model_II	0.9983	0.9928	0.9895	+0.0055
Dataset1	0.9983	0.9960	0.9895	+0.0023

Plus an 11-point sweep (single seed): quantum $>$ sham at **21 of 22** points, peak $\Delta=+0.112$ at $N=1500$.

But this is what the literature warns about

A favourable comparison against a **handicapped sham** and an **oversized ViT**. What about a plain CNN?

Evolution (2): the null baseline demolishes it

The missing control: a *plain CNN with direct readout* — same encoder, same protocol, **fewer** params (**93,763**, smallest in the program).

Bottleneck-tax ledger (Model-II, $N=500$):

Arm	AUC	vs. plain-lin
plain-lin (un-taxed CNN, 93.8k)	0.9129	—
QVF-scratch quantum (forced 8-dim readout)	0.8206	−0.092 (tax)
QVF-scratch sham (forced linear readout)	0.7683	−0.145
<i>recovery</i> (quantum — sham, in bottleneck)		+0.052
<i>net</i> (quantum — plain-lin)		−0.092

Reading

The 8-dim bottleneck is a **handicap, not a regulariser** — it loses even at $N=500$. The “21/22 win” was measured *inside* a family a plain CNN dominates.

Evolution (3): Wide is not enough $\Rightarrow$ Hybrid

QVF-Wide: readout 8 $\rightarrow$ 52 observables ($\langle Z \rangle, \langle X \rangle, \langle Y \rangle, \langle Z_i Z_j \rangle$):

N (Model.II)	wide-Q (52)	wide-sham	orig-Q (8)	plain-lin
500	0.8450	0.7850	0.8206	0.9129
3000	0.9631	0.9575	0.9808	0.9849
8000	0.9861	0.9906	0.9896	0.9942

Regresses at mid- N ; sham wins at $N=8000$. Width is not the fix — *forcing all 128 CNN dims through the circuit* is the tax.

QVF-Hybrid: the quantum branch in residual

$$\hat{y} = W_{\text{head}} \text{concat}(\text{LN}(\mathbf{h}), \text{LN}(\mathbf{m})) + b$$

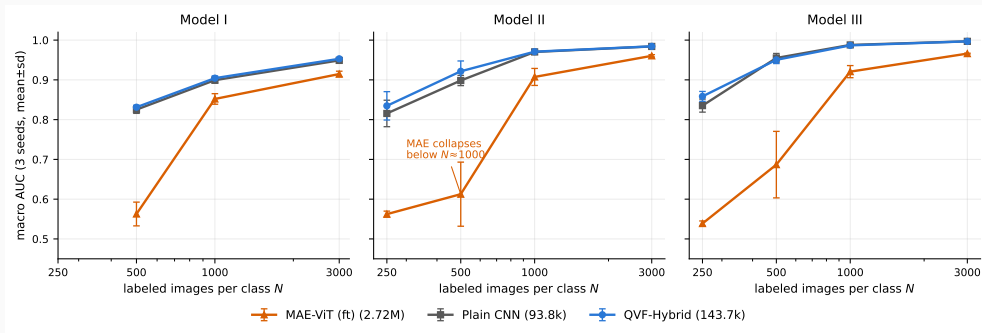
Zeroing the $\mathbf{m}$ -block reduces **exactly** to plain-lin (verified, error = 0.0) $\Rightarrow$ **plain-CNN floor holds by construction.**

The control protocol (fixed before the runs)

- **Capacity-matched sham.** Circuit $\rightarrow$ Linear+Tanh; sham always has *more* params (156,923 vs **143,655**).
- **Null baseline.** plain-lin (93,763), plain-mlp (142,837) — *smaller* than the quantum model.
- **Bit-identical subsampling.** Same RNG across methods $\Rightarrow$ every arm sees *exactly* the same images per seed.
- **Best-val-AUC** checkpoint, never the last epoch.
- **Multi-seed**, seed-*paired* (3 seeds; 10–20 for significance).
- **Practical bar:** $< 3\%$ AUC is not claimed as an effect.
- **Pre-registration:** success criteria fixed *before* launch.

MAE-ViT gets its *full* recipe: mask-0.9 pretraining on unlabeled no_sub, Adam 5×10^{-5} , 50 epochs, rot90/flip augmentation.

Main result — three datasets, three paradigms



Macro-OVR AUC vs. labeled samples/class, 3 seeds (mean $\pm$ s.d.). **QVF-Hybrid** (143.7k, no pretraining) tracks or leads the plain-CNN family everywhere; **MAE-ViT** (2.72M, full pretraining + aug + $2.5\times$ epochs) **collapses** below $N \approx 1000$.

The significance battle (two-stage, pre-registered)

Stage 1 — all cells, $n=10$, Holm over 6 tests:

Cell	contrast	pos	mean Δ	p raw	p Holm
Model.II $N=250$	Q>plain	7/10	+0.0153	0.032	0.161 n.s.
Model.II $N=250$	Q>sham	8/10	+0.0189	0.0098	0.059 <i>marginal</i>
Model.II $N=500$	Q>plain	7/10	+0.0096	0.077	0.309 n.s.
Model.II $N=500$	Q>sham	4/10	+0.0017	0.539	1.000 n.s.
Model.I $N=500$	Q>plain	6/10	+0.0009	0.237	0.712 n.s.
Model.I $N=500$	Q>sham	5/10	-0.0010	0.615	1.000 n.s.

Stage 2 — pre-registered sequential confirmation, $n=20$, $\alpha=0.01$:

Cell	contrast	pos	mean Δ [95% CI]	p	verdict
Model.II $N=250$	Q>sham	15/20	+0.0140 [+0.0048, +0.0233]	0.0032	significant
Model.II $N=250$	Q>plain	11/20	+0.0081 [-0.0010, +0.0172]	0.088	n.s.

Module-level significant vs. sham — but **sub-3%**, **single-cell**, and vs.-plain is **n.s.** Sequential design disclosed; α fixed before Stage 2.

Mechanism: entanglement is compensation, not information

Forced-readout: entanglement is load-bearing — **ENT > NOENT > SHAM** in 6/6 cells; at the peak cell (II, 1500) it carries $\approx 66\%$ of the quantum-over-sham margin.

Residual (hybrid), $n=10$:

N (Model.II)	ENT	NOENT (Rot only)	SHAM	plain-lin	ENT-NOENT
250	0.8339	0.8317	0.8149	0.8186	+0.0022 ($p=0.22$)
500	0.9198	0.9198	0.9181	0.9102	-0.0000 ($p=0.47$)

Data-processing inequality

$\mathbf{m} = g(\mathbf{h})$ is deterministic $\Rightarrow I(Y; (\mathbf{h}, \mathbf{m})) = I(Y; \mathbf{h})$: **the branch cannot add label information**, only reshape geometry.

Peak margin (II, 1500): +0.108 **forced** $\rightarrow$ +0.001 **hybrid** — a 100 $\times$ collapse. Entanglement mattered only while the architecture was crippled.

Paradigm duel: QVF-Hybrid vs. MAE-ViT

Data	N	QVF-Hybrid	MAE-ft	Δ	plain-lin	verdict
Model_II	250	0.8347	0.5623	+0.272	0.8155	PASS 3/3
Model_II	500	0.9216	0.6125	+0.309	0.8984	PASS 3/3
Model_II	1000	0.9709	0.9074	+0.064	0.9702	PASS 3/3
Model_II	3000	0.9843	0.9604	+0.024	0.9841	× (< 3%)
Model_I	500	0.8312	0.5627	+0.269	0.8252	PASS 3/3
Model_I	1000	0.9043	0.8520	+0.052	0.8993	PASS 3/3
Model_I	3000	0.9530	0.9147	+0.038	0.9490	PASS 3/3

Scoreboard: 10 of 11 cells across three datasets clear the pre-registered $\geq 3\%$ bar (+3.1 to +32.0 pp).

Significance: Holm $p=0.003$, 10/10, +22–28 pp on three low- N cells. **MAE-probe** (frozen + linear): 0.49–0.51, random-level.

Honest attribution

The win belongs to the **CNN inductive-bias family**, not the circuit — plain-lin also crushes MAE-ft (II, 500: 0.8984 vs 0.6125). QVF-Hybrid is the best *variant* of that family on Model_II.

Negative results (honesty is the product)

Lever	Outcome
Wide readout (52 obs.)	regresses at mid- N ; sham wins at $N=8000$
Qubit scaling $8 \rightarrow 12$	does not transfer : (II,500) identical (0.9216=0.9216); on Model_I the nq12 <i>sham</i> (0.8471) beats quantum (0.8387); params balloon to 639k
Circuit LR (10^{-2})	inert (0.9188 vs 0.9216)
TDA (persistent homology)	all 3 criteria failed ; the pairing-broken null <i>matches</i> real features $\Rightarrow$ no topological label information
QMAE (raw-pixel amplitude)	sham wins everywhere (-0.012 to -0.050) across $L=3 \rightarrow 20$, res $16 \rightarrow 64$, $N_Q=12 \rightarrow 14$
Angle enc., Model_III full	unstable (AUC oscillates 0.27–0.61)

The old “more qubits $\Rightarrow$ bigger quantum win” law was *bottleneck compensation*: remove the bottleneck and the lever dies — exactly what DPI predicts.

Generalization: a third dataset and a hard case

Model_III (3 seeds):

N	hyb-Q	hyb-sham	plain-lin	MAE-ft	Δ vs. MAE
250	0.8585	0.8314	0.8353	0.5387	+0.320
500	0.9504	0.9460	0.9551	0.6868	+0.264
1000	0.9867	0.9856	0.9879	0.9207	+0.066
3000	0.9968	0.9968	0.9969	0.9660	+0.031

At (III, 250): $Q > \text{sham} + 0.027$ (3/3 seed-paired) — low- N direction replicates on a third dataset ($n=3$, no significance claim). All four cells clear the $\geq 3\%$ paradigm bar.

CommonTest (hard case, 80 ep, full data): plain-mlp 0.8087 / plain-lin 0.7951 / hyb-S 0.8025 / hyb-Q 0.7968 / MAE-ft 0.6207.

- Paradigm ordering reproduces: CNN family $\gg$ MAE-ft.
- Hybrid floor holds; quantum $\approx$ sham — the low- N gain lives in a regime CommonTest never reaches.

Conclusion — the three-tier claim ladder

1. **Module** (circuit vs. matched sham): **statistically significant** at (II, 250), pre-registered two-stage design ($n=20$, $p=0.0032 < \alpha=0.01$). Magnitude +1.4% — **below our 3% practical bar**. Driven by amplitude-embedding structure, **not** entanglement.
2. **Model** (vs. plain CNN): **guaranteed floor by construction**; vs.-plain **not significant** ($p=0.088$) $\Rightarrow$ no model-level claim.
3. **Paradigm** (vs. MAE-ViT): **significant** (Holm $p=0.003$, 10/10, +22–28 pp), 10/11 cells over the bar across three datasets, 1/19 params, **zero** pretraining data — credit to the **CNN family**.

The honest headline

We do not claim quantum advantage. We claim a significance-tested *module-level* contribution too small to justify quantum hardware on this task — plus a mechanism (DPI) explaining why it cannot be larger, and the negative results that bound it.

Paper, code, figures, and every number:

github.com/leo07010/gsoc2026-deeplense-qml